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PHYSICS
Paper 1
APRIL, 2026

2 ½ hours

ST JAMES SECONDARY SCHOOL-KITAGOBWA
Uganda Certificate of Education

PHYSICS

Paper 1

Theory

2 Hours 30 Minutes

INSTRUCTIONS TO CANDIDATES:

- ✓ *This paper consists of* **two** *sections; A and B. It has* **seven** *examination items.*
- ✓ *Section* **A** *has* **three** *compulsory items.*
- ✓ *Section* **B** *has two parts; I and II. Answer* **one** *item from each part.*
- ✓ *Answer* **five** *items in all.*
- ✓ *Any additional item (s) answered will* **not** *be scored.*
- ✓ *All answers* **must** *be written in the answer booklet (s) provided.*

SECTION A

Ans*wer* **all** *items from this section* **Item**

1.

In a certain music concert that took place at night, a man played a guitar on a floating stage surrounded by disco lights flashing red, blue and green in the middle of the lake. The audience on the boats and shores were wearing yellow clothes with black spots on

them. The audience was surprised about the new appearance of the colours of their clothes. The sound waves from the guitar travel through the air with a frequency of $\mathbf{440}\boldsymbol{Hz}$. The organizers also projected laser light that travels through air with a frequency of $\mathbf{4.7 \times 10^8}$ M$\boldsymbol{Hz}$ illuminating waves on the lake surface to aid visibility. The shores of the lake were $\mathbf{15}\boldsymbol{m}$ away from a tall storage building. Boys **X** and **Y** standing in the same direction and in line with the playing music from the lake heard the sound at different intervals of time which attracted them to go and observe what was taking place at the lake. Boy **X** heard the sound after **3s** and boy **Y** heard the sound after **4s**. **Hint:** $Speed\ of\ sound\ in\ air = \mathbf{330}\boldsymbol{ms}^{-1}$

$Speed\ of\ light\ in\ air = \mathbf{3.0 \times 10^8}\boldsymbol{ms}^{-1}$

Task: As a physics student, help the two boys to understand; (a)
why they heard the sound at different intervals.

(b) why the colour of the clothes of the audience kept on changing when coloured lights flashed on them.

(c) why the laser source of light preferred to provide laser light that enhanced visibility late in the late hours of the night.

(d) Compare the wave length of the sound waves and laser light waves in air medium.

Item 2

A certain town experienced an atomic bomb blast, many people were killed in the incidence and others survived with severe injuries, the government advised those survived to re-locate to other places of the country because this village was not safe for them. The scientist carried out a scientific investigation using detectors of radiations and found out that the environment was still radioactive as shown in the table below.

Time (**days**)	0	22	42	62	82	102	122	142	162
Count rate (**min^{-1}**)	106	83	66	56	47	36	31	26	15

The scientists recommended that the environment will be safe for use again when obtained half-life value is less than 60 days and they will need like 1 year (**365 days**) for the environment to be free from radioactive wastes and materials.

Task: As a student of physics, use your knowledge to:

(a) Help the people to know the value of half-life and advise them accordingly.

(b) Sensitize the people about the risk associated with radioactive materials and how they should be handled.

(c) Help the people to understand the value of number of neutrons of a nuclide **Y** formed when a radioactive substance **X** of mass number **208** and atomic number **104** decays by emitting **3** alpha particles, **4** beta particles and gamma rays to form a nuclide **Y**.

Item 3

On a certain day at around **1:30pm** while in the dining hall, learners watched heavy fog-rains and floods being experienced in a certain outside country on an international T.V live channel. To worsen matters, the floods were happening at night and this risked many natives as many of them were ambushed while asleep. Learners wondered how it would

be night and seriously raining in an area yet it was day and the Sun was highly shinning at that time in their school.

Task: As a physics learner, help the learners clear their queries about;

(a) Occurrence of the floods in one area yet it was shining in their school at same time.

(b) Why it was night in that outside country yet it was day-time in their area?

(c) How T.V signals broadcast from where the floods were happening reached them.

SECTION B PART 1

Answer **one** *item from this part*

Item 4

On a construction site, a builder uses an electric lifter to transfer the mortar from the ground to a height of **17.0m** on the building. When power went off and work has to continue, the builders got a rope, a pail of negligible mass with a handle that lift a maximum of **20kg** of mortar. They want to design a simple machine with a help of a grooved rim which can enable them resume work. Given that the rope has a maximum thermal strength of **3.8kJ**.

Hint: Assume all the potential energy is converted to heat energy.

Task. As a physics learner,

(a) Help the builders to design the machine and guide the on how it works.

(b) Advise whether the rope will withstand the heat generated or it will break.

(c) Suggest the necessary factor to consider that enable the rope to success fully execute the work.

Item 5

Akello hotel has its bathrooms situated on the **2nd** floor of a building. A customer of the hotel expects to bathe water at **32°C**. The hotel provides **10** litres of water at **20°C** to each customer. A boiler on ground floor heats water to **72°C** for the customers to use. The hotel management does not allow its workers to carry the hot water via the staircase.

Task: Having studied physics;

(a) Help the hotel management to determine the quantity of hot water to be given to a customer for bathing.

(b) Advise the hotel management on how to keep the boiled water hot for a long period of time without keeping the boiler on.

(c) Explain to the management how the water from the boiler can reach the second floor safely.

Use: Density of water = **1000 kgm^{-3}**,

Specific heat capacity of water = **4200 J kg^{-1} K^{-1}**

Acceleration due to gravity = **10 ms^{-2}**

PART II

Answer **one** *item from this part.*

Item 6

In a certain city, there is a tall building in which there is a magnetic separator that sorts iron and steel metals from the scrap. One day there was a heavy downpour accompanied by flashes of light in the sky that set the building on fire. The police rescue team immediately came and extinguished the fire though the workers did not know the exact cause of the fire.

One of the prominent engineers in the city advised the owner to install the device that can arrest such occurrences on the building; but the owners did not have an idea of the device. As they were looking for the solutions, the magnetic separator broke down and the workers ran to the owner complaining about how the machine can no longer sort the metals effectively. The owner who had no idea of how the machine works called his mechanical engineer and explained the situation. The Engineer advised him to import it from Dubai and the machine operates on 240V with a fuse rated 5A. When he went to one of the wholesale shops dealing in machines, he found there a similar machine with a labelled Resistance of 50Ω and bought it.

Task: As a physics learner,

(a) help the workers understand the cause of the fire
(b) explain how the device to be used arrests the occurrence
(c) help the owner understand how the machine he bought works and comment on its effectiveness.

Item 7

The welders in a certain workshop are troubled with their tools being shocked by electricity from a generator of 240V. When they visited a technician, they were advised to wind a copper wire to 3000 turns around a soft iron ring on the receiving part so as to output 120V suitable for their work shop operations fixed in a box.
However, the welders seem bothered of how this will be of help.

Task: As a learner of physics,

(a) Explain to the welders how the above design will produce power corresponding to their consumption.

(b) Help the welder to determine the number of turns to be would on the output part of the device.

(c) Comment on how efficient the device is if the ratio of current output to current input is 1.5.

(d) Advise the welders on how to improve the efficiency of the device.

THERE IS NO GAIN WITHOUT PAIN

****THE END****